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Wildfire risk in a changing climate: Evaluating fire weather indices and their global patterns with CMIP6 multi-model projections

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ABSTRACT

This study investigates potential wildfire risks across different global warming scenarios through a comparative analysis of two prominent fire weather indices: the McArthur Forest Fire Danger Index (FFDI) and the Canadian Forest Fire Danger Index (FWI), leveraging the latest multi-model projections from the 6th phase of the Coupled Model Intercomparison Project (CMIP6). Utilizing the Extreme Gradient Boosting (XGBoost) algorithm and the Shapley value, we identify the impacts of meteorological variables on fire weather danger as represented by FFDI and FWI. Our findings reveal that under the Shared Socioeconomic Pathways (SSP) 5–8.5 high-emission scenario, both FFDI and FWI project significant intensification of fire weather, particularly in historically recognized high-risk hotspots, demonstrating robust inter-model consistency. Notably, the future projections of FFDI indicate the likely occurrence of wildfires with unprecedented severity. The comparative analysis using Shapley values highlights substantial regional and index-specific variations in the contribution of meteorological input variables to fire weather simulations. While these global patterns are generally retained as global warming leads to a systematic reinforcement of all variables, in-depth regional scale analyses further uncover a stark contrast of dominant factors controlling FFDI and FWI. These findings stimulate discussion on the potential adaptability and discrepancies of empirically derived fire models, highlighting the need for future research to advance fire weather modeling with enhanced flexibility and multi-factor consideration.

1. Introduction

Wildfire plays a crucial role in the Earth system, significantly impacting ecosystems, the carbon cycle, atmospheric composition, as well as human health, society, and the economy (Duane et al., 2021; Flannigan et al., 2009; Huang et al., 2015). Recent decades have witnessed a notable global increase in wildfire activity, characterized by more frequent events, extended seasons, and unprecedented scales, resulting in substantial environmental and economic repercussions (Abram et al., 2021; Hardie et al., 2022; Jolly et al., 2015; Sharples et al., 2016; Tang et al., 2021; van der Velde et al., 2021; Wang et al., 2021). However, establishing a definitive link between the recent intensification of wildfires and anthropogenic global warming remains challenging due to the complexity and interconnected nature of the factors driving wildfire activity (Krikken et al., 2021).

Accelerated warming is commonly recognized to result in additional heat, enhanced atmospheric moisture demand, and intensified hydrological cycle (Giorgi et al., 2011), which are postulated to exacerbate wildfire-conducive conditions, accounting for the observed trends in wildfire behaviors. Numerous studies substantiate the impact of global warming on wildfire intensification (Abram et al., 2021; Duane et al., 2021; Flannigan et al., 2009; Jain et al., 2024; Libonati et al., 2022; Westerling et al., 2006). Rising temperatures, for instance, have been linked to reduced dead fuel moisture content, increased fuel flammability, more active lightning events conducive to wildfire initiation, and earlier snowmelt extending the fire season in regions such as the western United States and southeastern Siberia (Flannigan et al., 2009; Kim et al., 2020; J.-S. Kim et al., 2022; Liu, 2017; Matthews, 2013; Romps et al., 2014; Westerling et al., 2006). Wetter winters with sufficient precipitation, as another example, are crucial for generating sufficient

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fuel, especially in dry regions (Balch et al., 2018; Flannigan et al., 2009; Meyn et al., 2007). Precipitation, on the other hand, has competing effects, potentially altering the distribution of burned areas under climate change in contrasting manners (S.-K. Kim et al., 2022; Kuhn-Régnier et al., 2021; O et al., 2020). While lack of rainfall can increase fuel flammability and fire likelihood, it may also reduce fuel availability through long-term moisture deficit, and vice versa. Wind pattern changes are also critical in regional wildfire exacerbation. Stronger, drier downslope winds, like California's Santa Ana Winds, dry out fuels and intensify combustion along the way, increasing both the severity and extent of recent wildfires (Billmire et al., 2014; Nauslar et al., 2018; Westerling et al., 2004).

Human-related factors contributing to wildfires are potentially controllable and preventable, such as the misuse of traditional practices like straw burning for grassland recovery, and the inadequate warning and management system (San-Miguel-Ayanz et al., 2012). In contrast, meteorological conditions conducive to fires cannot be locally or temporarily managed. This inherent uncontrollability of weather patterns, coupled with the emergence of more extreme events further complicates efforts to mitigate wildfire risks. As further warming due to increasing greenhouse gas (GHG) emissions is expected, the frequency and severity of wildfires are highly likely to increase correspondingly. Recent studies have conclusively indicated that the intensification of extreme fire-weather conditions threatens to surpass the capabilities of existing wildfire management practices in the coming decades (Abatzoglou and Williams, 2016; Carnicer et al., 2022; Duane et al., 2021; Rogers et al., 2020). Therefore, it is imperative to anticipate where and how severe wildfire risk will emerge in the changing climate.

However, an unambiguous assessment of future wildfire risk is extremely challenging. While there is a growing scientific consensus that wildfire risks will escalate with rising temperatures in the future, the extent of severity and the specific regional hotspots at greatest risk remain highly uncertain. If the main sources of uncertainty can be confined to climate factors, apart from direct human factors, they are largely attributed to the uncertainty embedded in climate projections and the limited accuracy of indices used to measure wildfire risk. All climate models inherently contain errors in representing the effects of global warming in response to increased GHG emissions, which vary from region to region and from one variable to another. Since individual models generate their own combined responses of global warming and background climate variability to wildfire, it is essential to employ a multi-model ensemble that can embrace a diverse range of manifestations of global warming's influence on wildfires, offering a more comprehensive assessment than relying on any single model.

How to accurately measure wildfire risk is another issue beyond the reliability of climate projections. The potential of wildfire risks may not be adequately represented in a simple combination of relevant meteorological variables. This is because the dominance of individual variables may not be consistent, varying across different regions and time due to the highly dynamic nature of both the climate system and wildfire behaviours. Additionally, the synergistic effects of these variables, alongside ecological and human elements, further complicate the straightforward prediction and assessment of wildfire risks (Abram et al., 2021). Currently, the majority of indices quantifying potential fire danger employ empirical models that integrate the compounding impacts of temperature, humidity, precipitation, and wind on fuel moisture and fire behavior (Dupuy et al., 2020; Flannigan et al., 2016). Efforts have been ongoing to refine these indices by improving the empirical fit to better capture the nonlinear relationships and regional variability (Dowdy et al., 2010; Jurečka et al., 2019). However, there is no single formula that universally captures the multifaceted nature of wildfire risk across all geographical and climatic conditions, necessitating the use of multiple indices for a more accurate and holistic evaluation.

In this study, wildfire risks are evaluated through a comparative analysis of two distinct wildfire indices based on multi-model ensembles from the 6th phase of the Coupled Model Intercomparison Project (CMIP6). The McArthur Forest Fire Danger Index (FFDI) and the Canadian Forest Fire Weather Index (FWI) are employed as proxies for fire weather risks, both being extensively adopted in climate research. These two indices were selected not only for their popularity but also because they require the same meteorological input data, allowing a fair comparison (Dowdy et al., 2010). The comparative analysis of FFDI and FWI uncovers critical similarities and differences, facilitating cross-validation and enhancing the reliability of identifying regions with consistent patterns of wildfire risk across both indices. It is noteworthy that the two indices only offer a rough estimation of fire weather danger based on dead fuel moisture content and basic meteorological variables. Certain processes that could contribute to fire extremes may not be well accounted for by FFDI or FWI, including synoptic processes like cold fronts, downslope windstorms, and pyroconvection, which are also likely impacted by climate change (Cai et al., 2022; Di Virgilio et al., 2019; Duane et al., 2021; Nauslar et al., 2018), warranting more in-depth investigation in future works.

To establish the credibility of the adopted climate data, we first validate the historical simulations against the European Forest Fire Information System (EFFIS) reanalysis fire weather indices. The EFFIS reanalysis assimilates a variety of observations and is considered a good proxy for real-world fire events. Subsequently, to investigate how different levels of global warming affect future wildfire risks, climate projections under two different SSP scenarios are adopted: SSP1-2.6 (low challenges for mitigation and adaptation with a radiative forcing of 2.6 W/m^2 by the end of 2100, hereafter referred to as SSP126) and SSP5-8.5 scenario (high vulnerability to climate impacts with a radiative forcing of 8.5 W/m^2 by the end of 2100, hereafter referred to as SSP585). The evaluation of the SSP126 and SSP585 projections identifies vulnerable hotspots across different future pathways and quantitatively demonstrates the reduced wildfire risks associated with the low-emission scenario.

A particular focus is placed on identifying dominant meteorological factors that significantly modulate wildfire risks measured by FFDI and FWI, through the application of a machine learning model, namely the Extreme Gradient Boosting (XGBoost; Chen and Guestrin, 2016) algorithm and the Shapley value (SHAP; Lundberg et al., 2020; Lundberg and Lee, 2017). This approach enables the quantification of the relative contribution from various meteorological factors to wildfire risk and the interdependence among them. It also helps to distinguish the varying sensitivity of FFDI and FWI to specific climate conditions. Collectively, this research offers new insights into the underlying mechanisms driving climate-related exacerbation of wildfire risks under different levels of global warming, enhancing the understanding of the predictability and adaptability of existing fire weather models, while suggesting potential improvements for future fire weather projections.

2. Data and methodology

2.1. Climate simulations and validation data

2.1.1. CMIP6 ensemble simulations

In this study, the CMIP6 multi-model daily simulations are applied to generate FFDI and FWI for the historical period of 1976–2005 and the future period of 2070–2099. The historical simulations enable the evaluation of the CMIP6 models' capabilities in reproducing the major characteristics of FFDI and FWI as well as inter-model variability, serving as the baseline for future climate change analysis. For the future period, climate projections forced by the SSP126 and SSP585 scenarios are selected to represent climate conditions under different levels of global warming. The variables used to calculate FFDI and FWI are daily maximum near-surface air temperature (TASMAX, in degrees Celsius), wind speed (WIND, in kilometres per hour), relative humidity (RH, in percentage), and 24-h accumulated precipitation (PR, in millimetres). Based on data availability, twelve models participating in the CMIP6 are selected for comparative analysis, as summarized in Table 1. These

Table 1
Summarization of the full names and acronyms of 12 selected climate models from the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6). The first ensemble member (r1i1p1f1) of each model is selected.

Full Names of Selected CMIP6 Models	Acronyms
The Australian Community Climate and Earth System Simulator Earth System Model Version 1.5 (ACCESS-ESM1-5)	ACCESS
The Canadian Earth System Model Version 5 (CanESM5)	CanESM
The second generation CMCC Earth System Model (CMCC-ESM2)	CMCC
The third generation Earth System Model (EC-Earth3)	EC3
The Flexible Global Ocean-Atmosphere-Land System Model Grid-Point Version 3 (FGOALS-g3)	FGOAL
The fourth generation Geophysical Fluid Dynamics Laboratory Earth System Model (GFDL-ESM4)	GFDL
The Institute for Numerical Mathematics Climate Model Version 5 (INM-CM5-0)	INM
The Institut Pierre-Simon Laplace Climate Model for Phase 6 of CMIP, Low Resolution (IPSL-CM6A-LR)	IPSL
The sixth version of the Model for Interdisciplinary Research on Climate (MIROC6)	MIROC
The Max Planck Institute for Meteorology Earth System Model Version 1.2, High Resolution (MPI-ESM1-2-HR)	MPI
The Meteorological Research Institute Earth System Model Version 2.0 (MRI-ESM2-0)	MRI
Norwegian Earth System Model Version 2, Low atmosphere-medium ocean Resolution (NorESM2-LM)	NorESM

models encompass a broad range of warming sensitivities in response to GHG forcings. As FFDI and FWI have different rating systems in practice, we apply the 95th percentile definition for a general extreme fire danger level to facilitate global analysis, presenting the 30-year-mean climatology of fire weather indices above their annual 95th percentiles in both historical validations and future simulations (Richardson et al., 2021). This approach approximates real-life wildfire hazards and allows for a more rational analysis of the regions that are most vulnerable to wildfire risk.

2.1.2. Reanalysis-based fire weather indices and GLDAS land-cover mask

EFFIS provides historical reanalysis data for fire weather indices, including FFDI and FWI, based on the historical weather simulations from the fifth generation ECMWF atmospheric reanalysis (ECMWF ERA5) (Copernicus Climate Change Service, Climate Data Store, 2019; Vitolo et al., 2020). The EFFIS reanalysis data are enhanced by an extensive set of quality-controlled observations and are proven to be reliable approximations for historical meteorological conditions conducive to the initiation, spread, and sustainability of fire events (Gallo et al., 2022; McElhinny et al., 2020; Vitolo et al., 2020). In this study, we validate the historical CMIP6 ensemble simulations against the EFFIS reanalysis to establish credibility for our future projections and analyses.

Regarding environmental differences, the combustible fuel of the barren or sparsely vegetated areas may not be well accounted for in general fire danger modeling. To exclude these areas and the ocean, land cover masks with 1-degree spatial resolution generated by the Global Land Data Assimilation System (GLDAS) are applied. The GLDAS produces the land-sea mask based on the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite-based sensors, in which grid boxes recognized as ocean are assigned a value of 0, while all other grid boxes defined as land receive a value of 1. For the vegetation type mask, the modified MODIS IGBP vegetation type dataset from the GLDAS NOAH model is utilized, where Barren or Sparsely Vegetated areas are marked with a value of 16. To facilitate the masking of the ocean and Barren or Sparsely Vegetated areas, all meteorological variables, two wildfire indices output, and historical reanalysis data are bilinearly interpolated into 1-degree grids.

2.2. Calculation of fire weather indices

2.2.1. McArthur Forest Fire Danger Index (FFDI)

FFDI (McArthur, 1967) predicts fire danger by combining the antecedent fuel availability, represented as the drought factor, with the short-term responses of fuel moisture to weather variations (Dowdy et al., 2009; Finkele et al., 2006; Holgate et al., 2017; Noble et al., 1980). It was first developed in the form of a circular slide meter from field measurements of eucalypt forests in Australia and was later regulated into an operational algorithm based on mathematical equations of best fit to the meter's data throughout Australia (Holgate et al., 2017; Noble et al., 1980). Prior to 2022, FFDI was applied for fire danger ratings by the Australian Bureau of Meteorology. Due to its various limitations in representing current vegetation, fuel types and fire behaviours, FFDI has since been replaced by the Australian Fire Danger Rating System (AFDRS) (Grootemaat et al., 2024; Hollis et al., 2024). While it is no longer in routine use, the FFDI remains a well-established, widely studied model that is still applied in recent research supporting fire modelling advancements (Allen et al., 2022; Kussul et al., 2023; Plucinski et al., 2023), and it is well-suited to uncovering the dominant meteorological factors for empirical fire weather simulations under our research design. The calculation based on Noble et al. (1980) is shown in Eqn (1):

$$FFDI = 2.0 \times e^{(-0.450+0.987lnDF-0.0345RH+0.0338TASMAX+0.0234WIND)} \quad (1)$$

where DF is the drought factor, TASMAX (°C) is the daily maximum near surface temperature, RH (%) is the near surface relative humidity, and WIND (km/h) is the daily mean near-surface wind speed.

The DF can be perceived as an estimation of the fraction of dry fuel that is available for burning, with its value ranging from 0 to 10. It takes into account both the short-term wetting of fine fuels from recent rainfall and the effects of long-term rainfall deficits (or excesses) (Finkele et al., 2006; Lucas, 2010). In this paper we adopt the improved formulas by Griffiths (1999) (Eqns (2) and (3)) and the limiting x-function by Finkele et al. (2006) (Eqn (4)) for the calculation of DF:

$$DF = min\left[10, 10.5\left(1 - e^{-\frac{SMD+30}{40}}\right)\frac{41x^2 + x}{40x^2 + x + 1}\right] \quad (2)$$

$$x = \begin{cases} \dfrac{N^{1.3}}{N^{1.3} + P - 2}, N \geq 1 \text{ and } P > 2\ mm \\ \dfrac{0.8^{1.3}}{0.8^{1.3} + P - 2}, N = 0 \text{ and } P > 2\ mm \\ 1, P < 2mm \end{cases} \quad (3)$$

$$x_{lim} = \begin{cases} \dfrac{1}{1 + 0.1135\ SMD}, SMD < 20 \\ \dfrac{75}{270.525 - 1.267\ SMD}, SMD \geq 20 \end{cases} \quad (4)$$

where SMD (mm) is the soil moisture deficit, x represents the influence of a significant rainfall event in the previous 20 days, P (mm) is the sum of rainfall within that event, and N is the number of days since the day with the largest daily precipitation in that event. According to Eqn (3), a significant rainfall event contains at least 2 mm of precipitation and returns the lowest x values within a 20-day window, which in turn lowers the DF, indicating the decrease of fuel availability under recent substantial rainfall (Finkele et al., 2006; Lucas, 2010). Regarding operational calculation, the limiting x-function is developed to prevent the DF from rising too quickly during extended dry periods following a significant rainfall event (Finkele et al., 2006). If x is greater than the associated x_{lim}, the x_{lim} will be used in its place. The calculated DF will eventually be limited to the value of 10, as DF greater than 10 can occur whether x or x_{lim} is applied (Finkele et al., 2006).

The soil moisture deficit (SMD) reflects the dryness of the soil by demonstrating the balance of soil moisture, and it can be estimated by the Keetch-Byram Drought Index (KBDI) (Keetch and Byram, 1968), the Mount's Soil Dryness Index (SDI) or satellite observations like the Soil Moisture Ocean Salinity (SMOS) and Antecedent precipitation index (API), depending on the agreed local practice (Holgate et al., 2017). In this paper, we applied the KBDI as the SMD representative to calculate the DF following the original estimation by McArthur (1967). The fundamental formula to calculate KBDI is shown in Eqn (5):

$$KBDI_t = KBDI_{t-1} + ET - P_{eff} \tag{5}$$

where KBDI (mm) of a given day t ($KBDI_t$) is derived from the KBDI of the previous day t-1 ($KBDI_{t-1}$), ET (mm) is the amount of water leaving the soil via transpiration or evaporation, named evapotranspiration, and P_{eff} is the amount of water infiltrating the soil, named effective precipitation (Finkele et al., 2006; Holgate et al., 2017). Under the KBDI scheme, ET can be computed by Eqn (6):

$$ET = \frac{(203.2 - KBDI_{t-1})(0.968e^{(-0.0875T_{max}+1.5552)} - 8.3)}{1 + 10.88e^{(-0.00173R_{annual})}} \times 10^{-3} \tag{6}$$

where $KBDI_{t-1}$ (mm) is the KBDI value of the previous day, T_{max} (°C) is the maximum temperature over the past 24 h, and R_{annual} (mm) is the average annual rainfall accounting for the vegetation cover (Finkele et al., 2006; Holgate et al., 2017). As for P_{eff}, it is defined by Eqn (7):

$$P_{eff} = rain - (interception, runoff) \tag{7}$$

where rain (mm) refers to the total rainfall over the previous 24 h, and the interception and runoff is approximated as a constant amount of 5 mm. Following Finkele et al. (2006), the default value of previous KBDI to initialize the calculation is set to 0 (Finkele et al., 2006). Besides, the maximum value of KBDI is restricted to 203.2 mm to accurately align with the maximum field capacity of the soil (8 inches) set in the original practice (Keetch and Byram, 1968).

2.2.2. Canadian Forest Fire Weather Index

The Canadian Forest Fire Weather Index (FWI) System is a subsystem of the Canadian Forest Fire Danger Rating System (CFFDRS). It was initially developed based on the impacts of weather variables on both fuel moisture conditions for the forest floor and generalized fire behaviours for a standard jack pine stand (Dowdy et al., 2009; Van Wagner, 1987). Due to its extensive application in fire-related climate research, the FWI has been modified into a continuous function of both latitude and time, and has shown good adaptability and predictability in global and regional scale analyses (Dowdy et al., 2009; Dupuy et al., 2020; Quilcaille et al., 2022; Van Wagner, 1987; Viegas et al., 1999).

The FWI system consists of six modules in total. The primary three are designed for three types of forest fuels with different drying rates, namely the Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC) and Drought Code (DC) (Van Wagner, 1974). The three primary fuel moisture codes in conjunction with the wind variable are then linked in pairs to form two intermediate indices: Initial Spread Index (ISI) and Buildup Index (BUI), which in turn are combined to produce the Fire Weather Index (FWI) (Van Wagner, 1974). Specifically, the three primary fuel moisture codes assume that the instantaneous drying rate of each fuel is proportional to its current free moisture content (Van Wagner, 1974). The FFMC represents the moisture content of litter and other hardened fine fuels in a forest stand; the DMC represents the moisture content of loosely compacted, decomposing organic matter; and the DC represents a deep layer of compacted organic matter (Van Wagner, 1974). Since the calculations of FFMC, DMC and DC are partially recursive and requrie each components' previous day's value, standard initial values (i.e., FFMC = 85, DMC = 6, and DC = 15) from Wang et al. (2015) were adopted in this study to facilitate general assessment across the globe (Wang et al., 2015). In terms of fire behaviour indices, the ISI integrates fuel moisture of fine dead fuels and surface wind speed to estimate the rate of spread without considering the effect of varying fuel quantities, while the BUI determines the total fuel available for the spreading fire (Van Wagner, 1974). The ISI is an exponential function which doubles the FWI for every increase of 19 km/h in wind speed (Van Wagner, 1974). As for the BUI constructed based on DMC and DC, a harmonic mean is applied to preserve the main effect of DMC, as DC is slow reacting by nature (Van Wagner, 1974). Finally, FWI combines the ISI and BUI to produce a unitless index of general fire intensity potential. FWI can be interpreted as the energy output rate per unit length of fire front, and higher FWI values indicate higher fire danger potential (Van Wagner, 1974).

2.3. Extreme Gradient Boosting and shapley value to characterize the dominant factors behind wildfire risk escalation

To comprehensively assess the relative contributions from meteorological variables to the two fire weather indices, Extreme Gradient Boosting (XGBoost) algorithm is first employed to build a prediction model to study the highly non-linear relationship between FFDI/FWI and the input features including temperature (TASMAX), drought factor/precipitation (DF/PR), wind speed (WIND), and relative humidity (RH). XGBoost models sequentially construct new models to predict residuals of prior nodes and eventually assemble all of them for prediction, which technique is known as gradient boosting (Chen and Guestrin, 2016). The XGBoost is selected to leverage its efficiency and power in regression tasks, as well as its superior interpretability compared to other deep-learning models, due to its tree-based structure that offers a more transparent decision-making process. For model training, 30-year time-series of FFDI/FWI and the corresponding input variables are used, which are extracted from the historical simulation ensemble (1976–2005), and the future simulation ensemble (2070–2099) under the SSP126 and SSP585 scenarios. The model exhibited satisfactory performance in both training and testing, as indicated in Table S2, confirming its reliability for further analysis.

We then employ the SHapley Additive exPlanations analysis technique (SHAP; Lundberg et al., 2020; Lundberg and Lee, 2017) to interpret the XGBoost model prediction process and measure the sensitivity of meteorological factors to fire weather indices. The SHAP analysis utilizes an additive feature attribution approach and formulates the model's output as a linear combination of its input variables (Abdollahi and Pradhan, 2023; Cheng et al., 2021; Qayyum et al., 2024a, 2024b), which is expected to accurately capture the contributions from meteorological variables to fire indices at both local and global scales. The Shapley value is based on game theory and represents the weighted average of the difference between estimation with and without the input feature of interest, which is formulated in Eqn (8). Basically, a negative Shapley value for one feature indicates that the inclusion of it lowers the predicted FFDI or FWI, while a positive Shapley value suggests that its inclusion increases the predicted FFDI or FWI.

$$\varphi_j = \sum_{S \subseteq \{x_1, x_2, x_3, \ldots, x_p\} \setminus \{x_j\}} \frac{|S|!(p - |S| - 1)!}{p!} \left(f_{X_S \cup \{x_j\}}(X_S \cup \{x_j\}) - f_{x_s}(X_s) \right) \tag{8}$$

where φ_j is the Shapley value of the jth input feature; $\{x_1, x_2, x_3, \ldots, x_p\}$ is the set of all input features; p is the number of all input features; $\{x_1, x_2, x_3, \ldots, x_p\} \setminus \{x_j\}$ is a set of all features without x_j; S is the subset of $\{x_1, x_2, x_3, \ldots, x_p\} \setminus \{x_j\}$, $f_{x_s}(X_s)$ is the prediction of feature subset S; and $f_{X_S \cup \{x_j\}}(X_S \cup \{x_j\})$ is the prediction of feature subset S with x_j.

It is important to note that the FFDI and FWI are calculated using meteorological variables as inputs, suggesting a predefined relationship between climate inputs and fire indices. However, this relationship is highly non-linear due to multiple calculation steps and parameterized equations that manually transform data and constrain it within

reasonable ranges. To quantify this relationship, data-driven analysis tools are essential. This methodology has been applied in index analysis by Zhou et al. (2024), using SHAP to study the impacts of temperature and relative humidity on the calculated heat index. Several studies in the existing literature (Abdollahi and Pradhan, 2023; Felsche and Ludwig, 2021; Zhang et al., 2023) have also employed SHAP to examine different factors' contributions to target variables, validating the feasibility of combining machine learning with SHAP values to investigate non-linearly related variables.

3. Results and discussion

3.1. Validation of fire weather indices from CMIP6 historical simulations

To assess the performance of the CMIP6 multi-model ensemble in reproducing historical fire weather danger, we compared the mean climatology of the historical top 5% fire weather indices simulated by the CMIP6 ensemble with that of the EFFIS reanalysis, as presented in Fig. 1. The spatial patterns of the simulated FFDI and FWI are generally in good agreement with those from the EFFIS reanalysis, both demonstrating high spatial correlation of approximately 0.91. Furthermore, the CMIP6 ensemble is capable of identifying historical wildfire hotspots, including regions in the western US, Brazil, Central and Southern Africa, western Australia, and Southwestern Asia. However, it tends to underestimate the values of fire danger indices for most of these regions, while overestimating for certain areas in Southwest Asia. Nonetheless, the global biases of the simulated indices remain within an acceptable range.

The observed discrepancy between the EFFIS reanalysis and the CMIP6 ensemble simulations may be attributed to several factors. One primary reason is that the EFFIS reanalysis incorporates historical observational calibrations, which enhances its representation of historical extreme fire events particularly in the wildfire hotspots, that may not be fully accounted for through empirically derived fire weather modeling (Vitolo et al., 2020). Although the CMIP6 models exhibit a considerable range of performance levels, the ensemble method can only provide a comprehensive representation of their simulation capabilities, which may also lead to underestimation in regional extremes. Another potential cause for the overestimation of both indices in Southwest Asia is the region's arid climate, characterized by low humidity levels and high temperatures (Cullen, 2005). These conditions create an environment that is conducive to extreme fire weather, as the occurrence of consecutive dry soil moisture anomalies is often observed prior to the largest wildfire, which may in turn increase the likelihood of overestimated simulations (O et al., 2020).

Considering index differences, the FFDI demonstrates a more extensive range of relative underestimations compared to FWI. In addition, the CMIP6 ensemble simulation of FWI demonstrates superior performance in high latitudes when compared to FFDI. This discrepancy can be attributed to the regional suitability of empirically derived models, as FWI is customized for use in higher latitudes than FFDI. Furthermore, the FWI possesses design features with the potential for application beyond local fuel types, such as specifying the impacts of moisture content on different fuel types (Hollis et al., 2024; Van Wagner, 1987). In contrast, the FFDI model has been demonstrated to be inadequate for representing fire danger in vegetation types that are structurally dissimilar to forest or grass, such as shrublands and spinifex (Anderson et al., 2015; Burrows et al., 2018; Grootemaat et al., 2024; Hollis et al., 2024). Due to this deficiency, FFDI has recently been retired from real-time operational use (Grootemaat et al., 2024; Hollis et al., 2024), but it is still being used for research purposes (e.g., Allen et al., 2022; Kussul et al., 2023; Plucinski et al., 2023).

For a more quantitative assessment of the capabilities of CMIP6 models, Fig. 2 presents the Taylor diagrams of FFDI and FWI corresponding to the upper 5 percent values during the historical period from individual model outputs, as well as their ensemble mean. The Taylor diagram is a useful graphical plot that simultaneously displays the spatial correlation coefficients, relative standard deviations, and their centered root-mean-square errors between the simulated and observed values. In a Taylor diagram, the better performance is indicated by a smaller root-mean-square error closer to 0, and with values of the spatial correlation coefficient and relative standard deviation closer to 1.

According to Fig. 2, historical simulations from twelve selected CMIP6 models have good spatial correlations with the reanalysis data, ranging from 0.82 to 0.93, while the performances are more concentrated for FWI compared to FFDI. It is worth noting that the CMIP6 multi-model ensemble outperforms any single-model simulation, with spatial correlations reaching 0.93 and a normalized standard deviation of 0.9 for both FFDI and FWI. Regarding single model performance, the MPI model emerges as the top performer with spatial correlations of 0.93 and 0.92 and a normalized standard deviation of 0.93 for FFDI and FWI, respectively. Following that, the simulation results of ACCESS and EC3 are consistently good across all models, with spatial correlations ranging from 0.91 to 0.92 for both indices. Overall, the CMIP6 multi-model ensemble generally outperforms single-model simulations in accurately reproducing historical fire weather indices, especially for the mean climatology of values above the yearly 95th percentile indicating the highest fire weather danger. This suggests that the ensemble approach can provide a more robust and comprehensive representation for the generation of prospective fire weather indices.

3.2. Future projections of fire weather indices under different warming scenarios

In this section, we focus on the future projections of fire weather indices and the changes in fire danger ratings under different warming

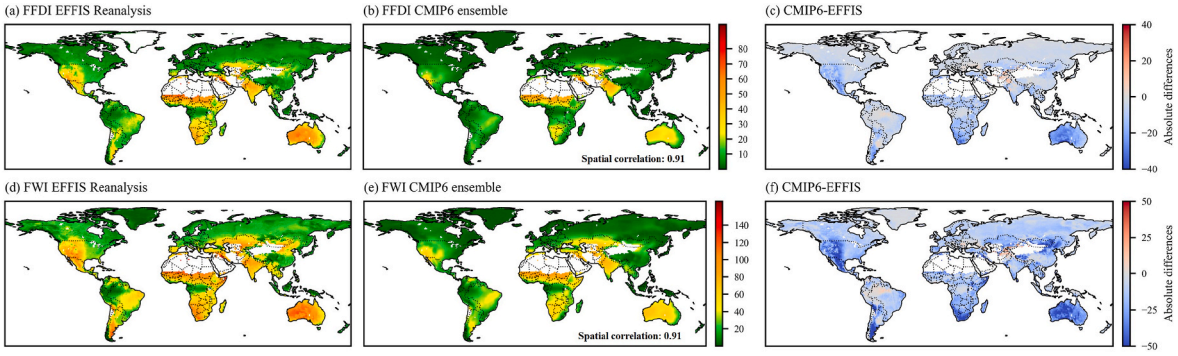


Fig. 1. Global distributions of FFDI and FWI above their 95th percentiles during the historical period (1976–2005) based on the EFFIS reanalysis data (a, d), the CMIP6 ensemble simulations (b, e), and the absolute differences between the two (c, f). The spatial correlation between EFFIS and CMIP6 ensemble is indicated at the lower right corner of (b) and (e).

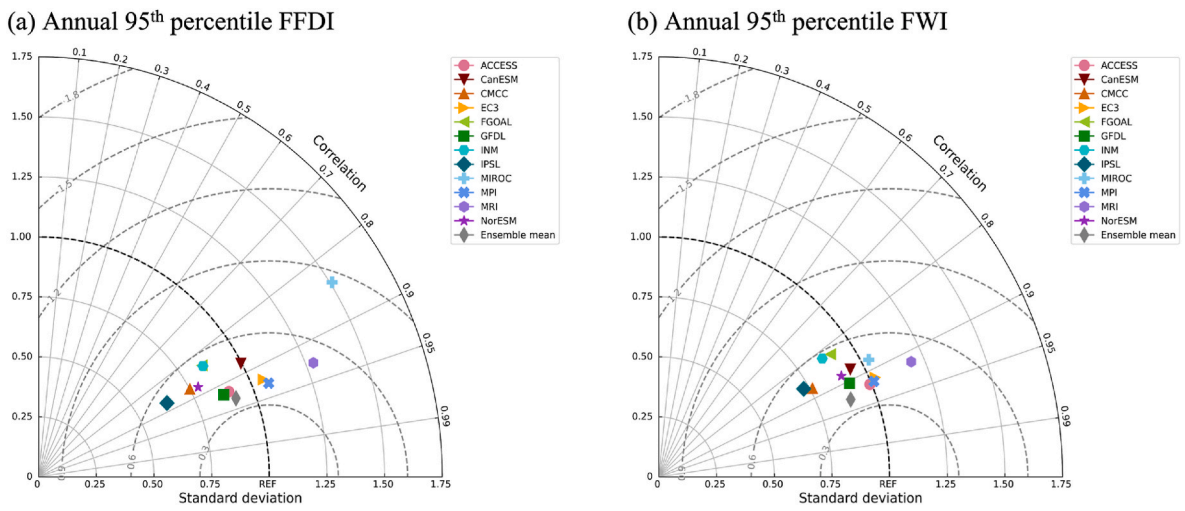


Fig. 2. Taylor diagrams of FFDI (a) and FWI (b) above their 95th percentiles during the historical period (1976–2005) based on selected CMIP6 models and their ensemble mean.

scenarios. When presenting future trends, Student's *t*-test is employed to assess the significance of changes, and a robust signal is defined when at least ten out of twelve models showed consistent and significant changes, which is commonly adopted in existing studies (Ha et al., 2023; McSweeney and Jones, 2013). Fig. 3 displays the change patterns of the 30-year mean FFDI and FWI climatology under the SSP126 and SSP585 scenarios, while Fig. 4 shows that for the climatology of the upper 5 percent of fire weather indices simulations respectively. Both figures use stippling to highlight regions with high confidence in the change patterns, where ten or more models consistently display the same positive or negative trends and pass the *t*-test.

According to Figs. 3 and 4, elevated fire weather indices are generally simulated in the stippled areas with consistent patterns of change, while the SSP585 scenario shows more significant increases than the SSP126 scenario. By the end of this century, the 30-year mean of FFDI is projected to be 43% and 15% higher under SSP585 and SSP126 respectively across all stippled areas. The mean yearly upper 5 percent of FFDI is expected to increase even more by an average of 49% under SSP585 and 17% under SSP126. Similarly, the 30-year mean of FWI is predicted to increase by 63% and 20% under SSP585 and SSP126, respectively. Meanwhile, the yearly upper 5 percent of FWI experiences increases of 52% and 17% under SSP585 and SSP126. The CMIP6 ensemble simulations of FWI and FFDI project consistent future change patterns, with the upper half of South America, the western portion of Asia connecting to the European region, South Africa, the west-central region of North America, and the western portion of Australia identified as hotspots of fire weather intensification by the end of this century, particularly under SSP585.

To compare the changing fire weather risks under different emission scenarios, we categorized the simulated fire weather indices into different fire danger levels and presented their frequency distribution as Fig. 5. The fire danger classification of FFDI follows Holgate et al. (2017), which comprises six levels from 'low' to 'catastrophic' (Holgate et al., 2017). Specifically, the 'catastrophic' category was introduced following the 2009 Black Saturday bushfires in Victoria, Australia to account for the exacerbated wildfire levels (Cruz et al., 2012). For FWI, the EFFIS categorization of seven levels is adopted, including the amended level of 'Very Extreme' for FWI values over 70 to facilitate

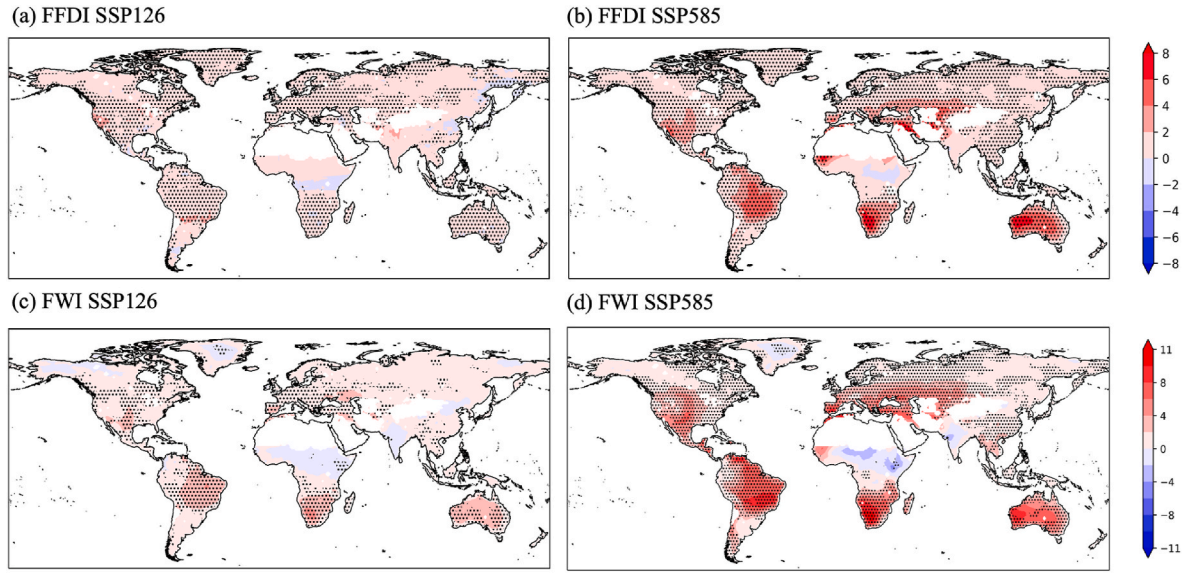


Fig. 3. Projected changes in ensemble mean of FFDI (a, b) and FWI (c, d) under the SSP126 and SSP585 scenarios (2070–2099) with respect to the historical baseline (1976–2005). Areas where at least 10 out of 12 models agree on the direction of change (positive or negative) and the variations have passed the Student's *t*-test for statistical significance are highlighted with stippling.

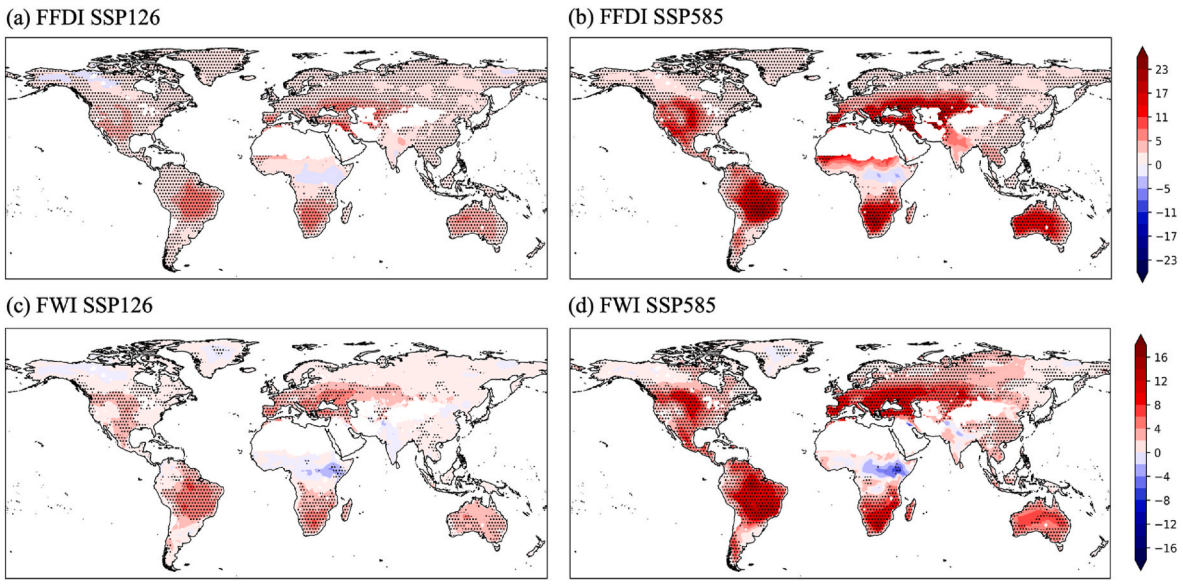


Fig. 4. As in Fig. 3, but for projected changes in ensemble mean of fire weather indices (FFDI or FWI) above their annual 95th percentile.

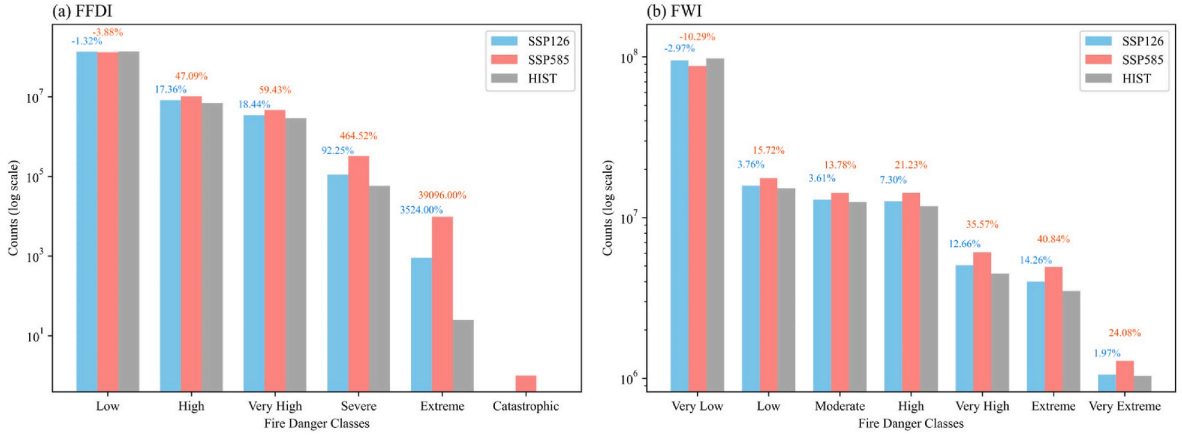


Fig. 5. Frequency distribution of simulated fire danger classes in the historical period (1976–2005) and future periods under the SSP126 and SSP585 scenarios (2070–2099). Panel (a) represents FFDI while panel (b) represents FWI. The y axis is set in logarithmic scale to show the counts for all grid points statistics during the studied periods, and the x axis indicates general fire danger ratings for FFDI and FWI. The blue and red boxes represent counts under the SSP126 and SSP585 scenarios respectively, and the gray box represents the historical counts. The numbers above the boxes with the corresponding colors represent the percentage changes under SSP126 (blue) and SSP585 (red) scenarios with respect to the historical simulations. In the case of FFDI (a), simulations of Catastrophic level only occur in the SSP585 scenario, thus no percentage change compared to historical scenario is available under the category of catastrophic.

more extensive investigation Copernicus Climate Change Service, Climate Data Store, 2019. The detailed thresholds of these fire danger ratings are supplemented in Table S1.

Overall, under the warming climate, increases in the number of days with higher fire weather dangers are expected by the end of this century according to FFDI and FWI projections. For both indices, the counts of days with a very low fire danger rating drop by 1%–10% compared with the historical scenario. Conversely, the numbers of days with a higher fire danger level increase steadily, with the greatest increases observed in the extreme categories. As a result, the lengthening of the fire season or expansion of fire regimes could occur under such a change pattern. Moreover, the elevation in the frequency of higher fire danger levels is more pronounced under the SSP585 scenario, and under the FFDI scheme. Specifically, for FWI, decreases of 10% and 3% are observed in the counts of days with very low fire danger under SSP585 and SSP126, respectively. Similarly, the FFDI model shows decreases of 4% and 1% under SSP585 and SSP126. As we consider all grid point statistics during the studied period, the days falling into the historically low fire danger category are at the ten to the eighth power level of base, which means a greater number of days are now categorized as having a higher fire weather risk, particularly in the SSP585 scenario. For the FWI category of low to high fire danger, date counts increase by 14%–21% under SSP585 and 4%–7% under SSP126. In FWI category of very high to extreme fire danger, counts rise by 36%–37% under SSP585 and 11%–13% under SSP126. As for FFDI, the categories of high and very high fire danger experience increases in occurrences of 47%–59% under SSP585 and 17%–18% under SSP126. Regarding the FFDI categories of severe and extreme, there are even more dramatic rises of approximately 5–391 times under SSP585 and 1 to 35 times under SSP126. It is worth noting that simulations falling into the catastrophic category of FFDI only occur under SSP585, thus no percentage change compared to the historical scenario is available under the category of catastrophic.

3.3. Comparative assessment of input meteorological variables' contribution to fire indices

Although FFDI and FWI project a comparable level of escalation in fire risks under the two future scenarios, the way those input meteorological factors contribute to their values differs greatly. Fig. 6 shows the spatial distribution of the major contributing variable to each fire index (i.e., the one with the highest mean absolute SHAP value in each grid) over the global land, which exhibits significant heterogeneity across various regions and for different indices. For FWI, the dominant meteorological factors are often arranged in latitudinal bands: in the northern hemisphere, WIND strongly influences the fire index on Greenland, likely due to its exponential form in the FWI algorithm, followed by TASMAX and RH in the vast majority of the land in the mid-latitudes and near the equator. Meanwhile, in the southern hemisphere, RH has a predominant impact on FWI, with TASMAX becoming influential in the high latitudes. As for FFDI, again, TASMAX and RH have widespread impacts across the global landscape. Nonetheless, DF, encapsulating moisture deficit and antecedent major rainfall events influences, exerts considerable impact in several spots in northeastern South America, Southern Africa, Northern Australia, and Southeastern Asia, the majority of which are tropical forests (Food and Agriculture Organization of the United Nations, 2020). DF is powerful in perceiving dryness conditions, particularly in regions experiencing high volume of rainfall regularly (Hadisuwito and Hassan, 2021; Plucinski et al., 2023), which could partially explain the high sensitivity of FFDI to it in these areas.

Overall, TASMAX and RH are identified as the two dominant meteorological variables in determining global fire risks, in line with the existing literature in global (Christian et al., 2023; Grillakis et al., 2022) and regional scales (Fan et al., 2023). In the continuously warming future, the rising temperatures and increased variability of relative humidity (Fan et al., 2023) are expected to have enhanced impacts on escalation of the fire risks (Christian et al., 2021). Interestingly, under future climate change, the spatial distribution of dominant factors remains largely unchanged (Fig. 6), which suggests that the contributing mechanism could be largely dependent on regional climate and the empirical function of fire danger index. It is noteworthy that under the FWI scheme, the PR-dominated regions are nearly indistinguishable based on the maps, although its relative contribution over the globe ranges from 28.6% to 58.7%. This might be due to the inherent complexity of PR's effect on estimated fire danger, which obfuscates its SHAP value patterns. Intermediate level of precipitation in winter/-spring could be pivotal to fuel availability and thus increase the fire danger in dry seasons, as observed in previous research where it corresponds to large burned area simulations (Archibald et al., 2009; S.-K. Kim et al., 2022; Koutsias et al., 2013). Meanwhile, for more humid regions, increases in precipitation could interrupt the accumulated dryness and flammability, mitigating fire danger estimations (Moritz et al., 2012).

Apart from identifying the dominant contributors globally, it is also essential to understand how different variables affect fire indices at finer regional scales. This will be explored through single-point analyses over five wildfire hotspots, the Amazonia, Southern New South Wales (SNSW) in Australia, Western Central Thailand (WCThai), California in the US, and Mulopwe in South Africa. Fig. 7 illustrates the mean absolute SHAP value of input variables over the five different regions and under different scenarios, estimating the potential impact of input meteorological variables on the fire indices. The results again reveal

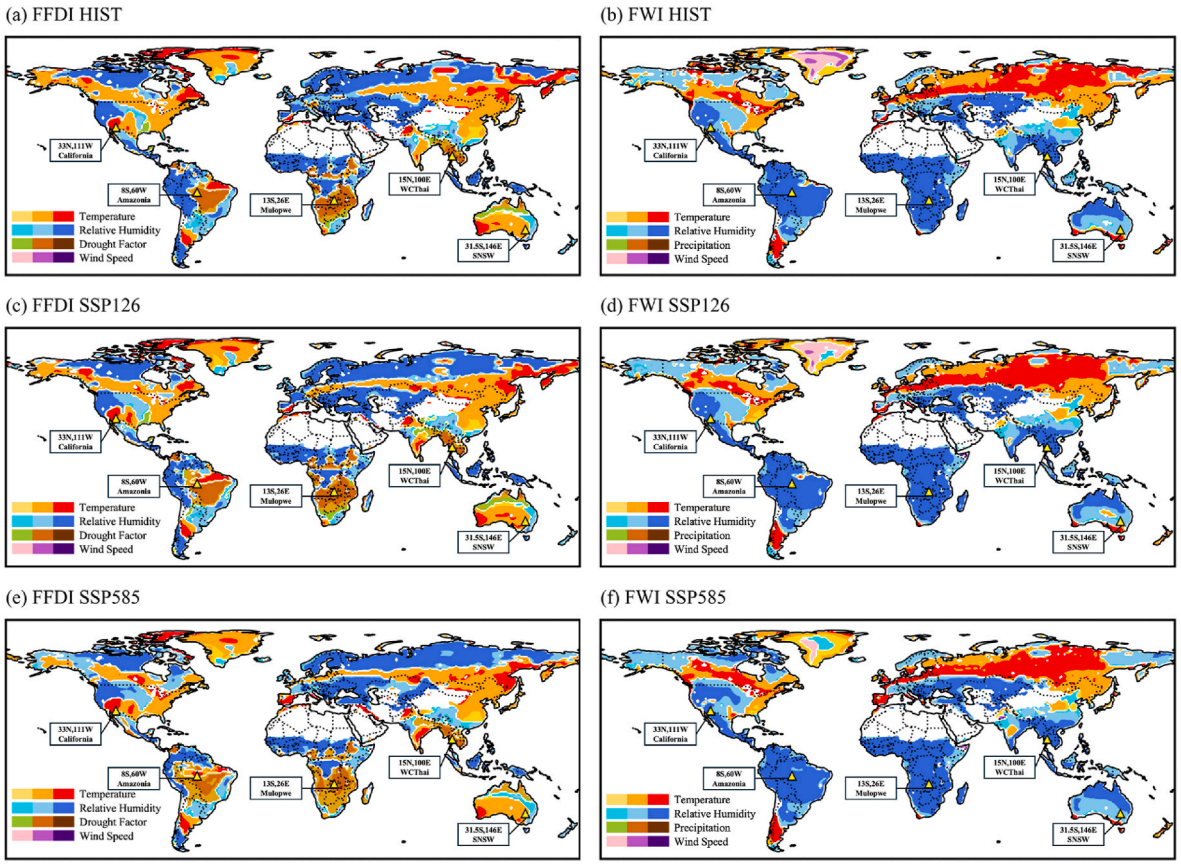


Fig. 6. Global distribution of dominant meteorological factors to the simulation of FFDI (a, c, e) and FWI (b, d, f), based on standardized mean SHAP value in the historical period (HIST; 1976–2005) and future periods under the SSP126 and SSP585 scenarios (2070–2099). Different series of colors represent different dominating meteorological factors, with the higher contribution represented by the darker shades. Five wildfire hotspots selected for single-point analysis are marked by yellow triangles with corresponding labels of their (abbreviated) names and coordinates.

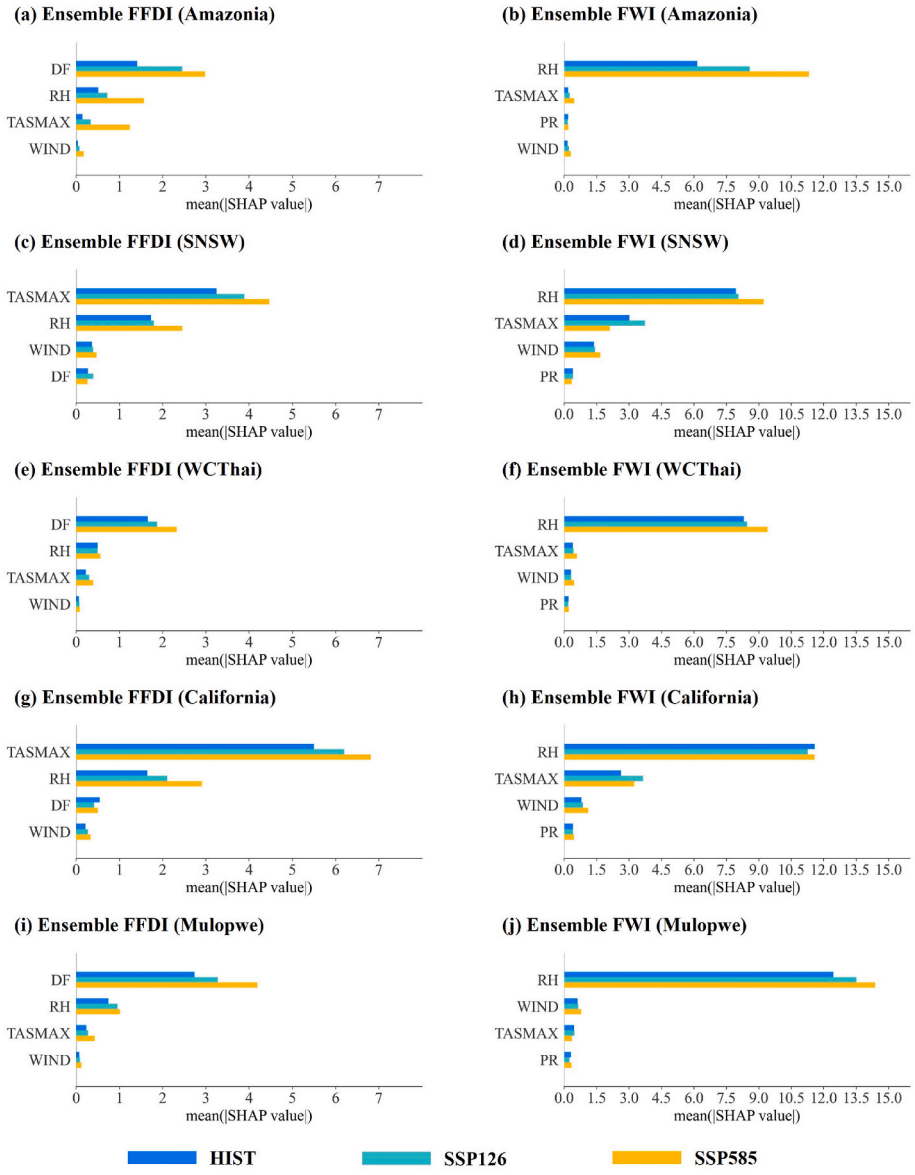


Fig. 7. Single point analysis of the relative contribution from daily maximum temperature (TASMAX), relative humidity (RH), drought factor (DF)/precipitation rate (PR), and surface wind speed (WIND) to the FFDI (a, c, e, g, i) and FWI (b, d, f, h, j) during the historical periods (HIST; 1976–2005) and future periods under the SSP126 and SSP585 scenarios (2070–2099), with respect to single grid points in Amazonia (8S, 60W; a, b), Southern New South Wales (SNSW; 31.8S, 145.6E; c, d), Western Central Thailand (15N, 100E; e, f), California (33N, 111W; g, h), and Mulopwe (13S, 26E; i, j).

significant regional and index-specific variability in the contributions of input variables. A distinct pattern emerges when examining the contribution mechanisms, which appear to be strongly influenced by the hydroclimatic characteristics of each region. Specifically, there is a clear demarcation between the behaviors observed in arid and humid environments.

Under the FFDI scheme, DF emerges as a primary driver in wet regions like Amazonia, WCThai, or Mulopwe, while playing a minimal role in arid regions including SNSW and California, where TASMAX appears more pivotal. This stark contrast could be attributed to the heightened sensitivity of humid regions to anomalous dryness (i.e., high DF) compared to arid ones (O et al., 2020; Plucinski et al., 2023). Meanwhile, RH acts as the predominant variable in affecting FWI. It almost exclusively determines the FWI in wet regions, while also possessing leading impacts in dry regions. This is likely because FWI is highly susceptible to short-term response (drying) of fuel moisture due to variations in RH, especially in the forests (Grillakis et al., 2022). While both FFDI and FWI concur that temperature and dryness are crucial factors in fire weather risk assessment, FWI appears to place greater emphasis on aridity. Under the global warming scenarios, almost all variables' contribution to fire indices are enhanced (mostly SSP585 > SSP126), resulting in elevated fire risks in a warmer future.

The impact of input variables on fire index calculations is rarely isolated; they interact, creating complex dynamics that influence fire risk (Koutsias et al., 2013). To evaluate these interactions and their impacts on fire indices, SHAP dependency plots are employed. Consistent with previous analyses, the contribution mechanisms exhibit distinct patterns between relatively humid regions (Mulopwe, WCThai, Amazonia) and drier areas (California, SNSW), with similar trends observed within each climate type, confirming the ability of SHAP in capturing the differences in regional climate. To maintain clarity and manage the figure size, the analysis results based on the two most representative cases: Amazonia (for humid regions) and SNSW (for dry regions) will be presented in Figs. 8 and 9. Dependency plots for California, WCThai, and Mulopwe are included in the Supplementary Information (Figs. S2 and S3) to provide a holistic view of conditions in

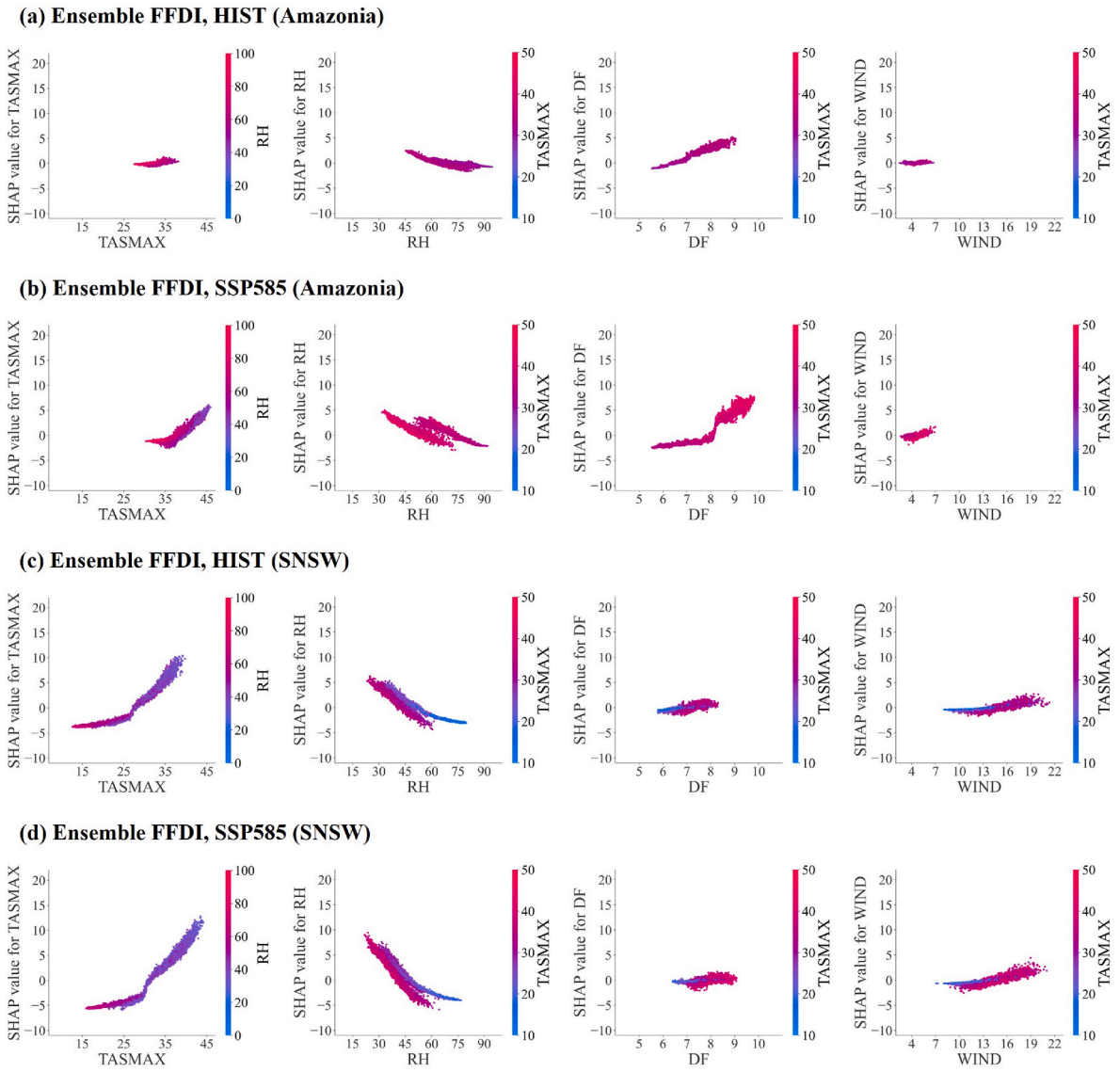


Fig. 8. Dependency plots of contribution from daily maximum temperature (TASMAX; in degrees Celsius), relative humidity (RH; in %), drought factor (DF; unitless), and surface wind speed (WIND; in km/h) to the McArthur Forest Fire Danger Index (FFDI) during the historical period (HIST; 1976–2005) and future periods under the SSP585 scenario (2070–2099) for single grid points in Amazonia (8S, 60W; a, b) and Southern New South Wales (SNSW; 31.8S, 145.6E; c, d).

different regions.

Overall, the underlying mechanisms of how these factors contribute to fire weather risks are similar under the two fire weather indices: Lower RH, stronger DF, higher TASMAX and WIND are conducive to higher estimated risks of fire weather, which concurs with common physical understanding. The dependency plots shed light on the rationale behind the major contribution from certain meteorological variables in Fig. 7. For example, the steeper slopes of the RH-SHAP and DF-SHAP scatters in Amazonia, the TASMAX-SHAP and RH-SHAP scatters in dry regions (Fig. 8) explain their high impacts on FFDI (Fig. 7). On the other hand, the pronounced steepness of RH-SHAP scatters in both humid/dry locations (Fig. 9) demonstrates FWI's heightened sensitivity to variations in RH, thus accounting for its dominant contribution (Fig. 7).

Consistent with our findings in Fig. 7, the dependency plots reveal an expanded range of meteorological variables' impacts under future warming scenarios, indicating elevated fire danger risks. The changes are more pronounced under SSP585 than SSP126 (Fig. S1), emphasizing the heightened wildfire risks under the high-emission scenario. In humid regions (Mulopwe, WCThai, Amazonia), FFDI exhibits a strengthened response to TASMAX under the SSP585 scenario compared to the historical period, with the scatters more tilted to the upper right (Fig. 8, Fig. S2), resulting in an increase in the mean absolute SHAP value (Fig. 7). Conversely, in dry regions (California, SNSW), the global warming effect on FFDI's sensitivity to TASMAX is less pronounced. Scatter points shift horizontally to the right and slightly extend at both ends (Fig. 8, Fig. S3), leading to a moderate increase in the mean absolute SHAP value (Fig. 7). FWI, however, shows consistent insensitivity to TASMAX across all regions and scenarios, with only a right shift of the patterns and no significant change in SHAP values. These results underscore that future fire danger estimations are expected to intensify under climate change, primarily due to increased temperatures. This enhancement is more evident through the FFDI scheme than FWI.

The interdependencies among meteorological variables, particularly with temperature, are evident under both FFDI and FWI, which is demonstrated by the data points' colors partitioned by different levels of hotness. Understanding these interdependencies is crucial as it suggests that besides the inherent changes in the variables under future climate change, the rising temperature could also critically affect their patterns and lead to a compounded alteration of their impact on the fire risk

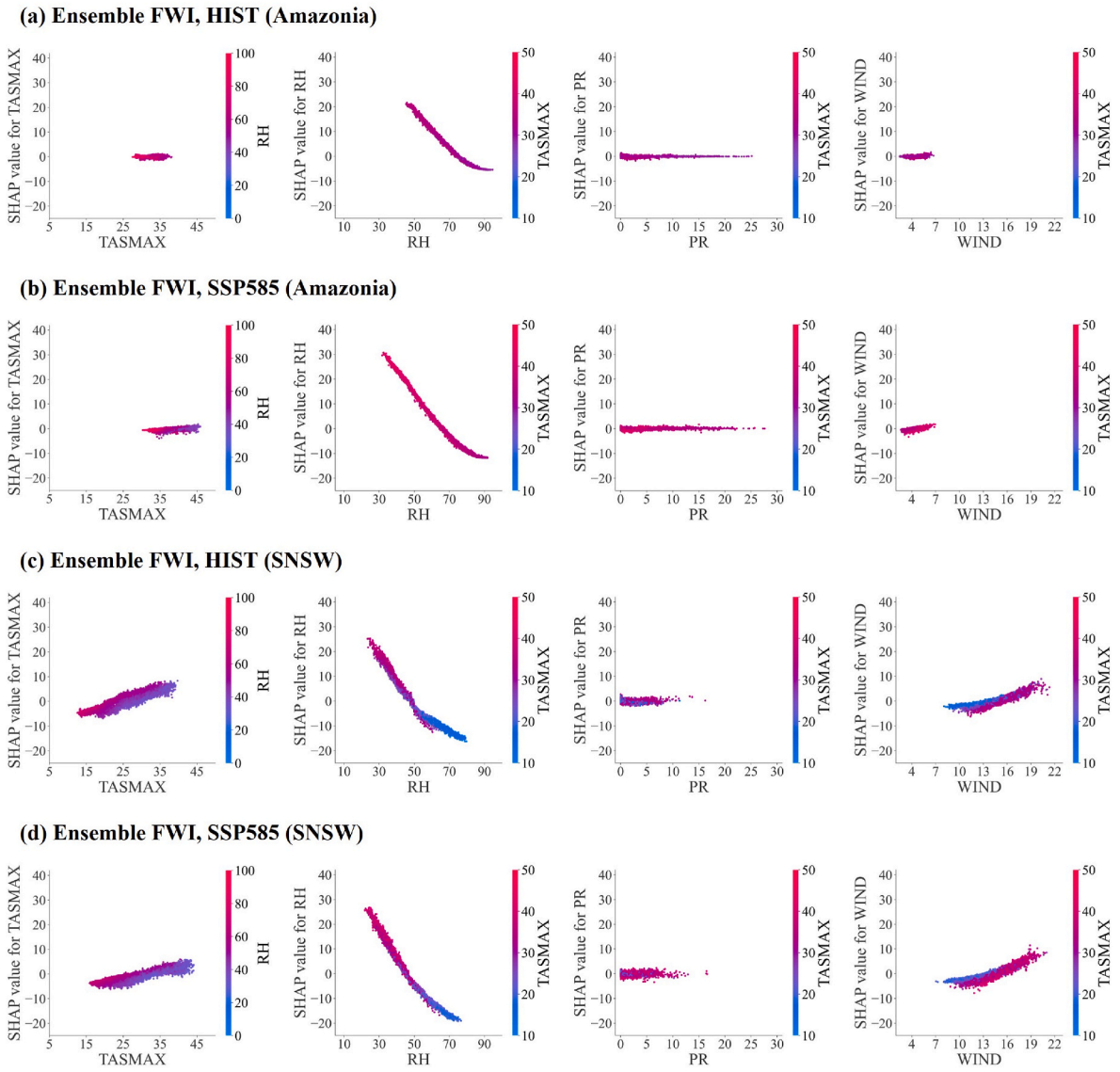


Fig. 9. Dependency plots of contribution from daily maximum temperature (TASMAX; in degrees Celsius), relative humidity (RH; in %), precipitation rate (PR; in mm/day), and surface wind speed (WIND; in km/h) to the Canadian Forest Fire Danger Index (FWI) during the historical period (HIST; 1976–2005) and future periods under the SSP585 scenario (2070–2099) for single grid points in Amazonia (8S, 60W; a, b) and Southern New South Wales (SNSW; 31.8S, 145.6E; c, d).

estimation.

For FFDI (Fig. 8, Fig. S2), the RH-SHAP points are distinctly divided into two color zones by TASMAX. This division is more pronounced in dry regions compared to humid areas, likely attributable to the greater temperature variability of more arid environments. Under future warming scenarios, the RH-SHAP pattern exhibits a leftward shift and extension at the positive end, indicating an increased positive contribution of RH to estimated fire danger. Similarly, the Drought Factor (DF) displays an extended right tail coupled with high TASMAX under SSP585. This change is particularly significant for Amazonia, WCThai, and Mulopwe, highlighting DF's enhanced impact on fire risks in these humid regions. As for FWI (Fig. 9), its most dominant factor, RH, exhibits a leftward extension in the overall patterns of SSP585, correlating with higher TASMAX. This trend suggests a potential future drying in the humid regions amidst global warming. Such a drying trend amplifies the positive contribution of low RH to fire danger estimation, underscoring these regions' heightened vulnerability to fire risk under future climate scenarios. This increased fire danger in humid regions aligns with findings from previous studies (Cochrane and Barber, 2009; Marengo et al., 2018; Talukdar et al., 2024), further underscoring the importance of addressing fire risk in vulnerable humid ecosystems as climate change progresses.

Even though the impact of input variables on fire indices could be reinforced due to enhanced range of variability under global warming scenarios, the overall patterns are still largely restrained by the empirical model and dependent on the regional climate (Figs. 8 and 9). Under the same structure of empirical formula, the contribution from different variables is not markedly affected by changes in variable magnitudes. This has profound implications for the formulation of fire risk assessment: to more accurately and comprehensively account for the temporal variation of fire risks, empirical formulas should be dynamically refined to incorporate regional climate uniqueness as well as non-linear temporal changes in the relationship between meteorological factors and fire weather index.

4. Summary and discussion

This study investigates the global wildfire risks under different levels of warming by computing two fire weather indices, FFDI and FWI, using CMIP6 multi-model simulations for both the historical period

(1976–2005) and the future period (2070–2099). To differentiate the future projections of FFDI and FWI and their sensitivity to climatic conditions, a novel approach based on the machine learning model XGBoost and the Shapley value is employed to account for the contributions of input meteorological variables to fire weather predictions. Results show that the historical CMIP6 ensemble simulations generally correlate well with the reanalysis-based fire weather indices, although they tend to underestimate the historical extremes for high-risk areas. For future projections, the FFDI and FWI consistently reveal significant intensification, especially in the wildfire hotspots, with dramatic increases in the occurrences of higher wildfire danger levels. This pattern is particularly dominant under the SSP585 scenario, which indicates higher vulnerabilities under climate change. Despite the consistent increase in fire risks projected by FFDI and FWI, the contributions of individual meteorological variables tend to diverge for the two indices, exhibiting large variability across different regions. Across the global landscape, temperature and relative humidity play predominant roles in affecting fire risk estimates. Meanwhile, regional investigations at five designated locations across Amazonia, Southern New South Wales, WCThai, California, and Mulopwe further offer refined details of the interdependence among the impacts of meteorological factors on fire risk escalations. Notably, under the global warming scenarios, the contributions from nearly all meteorological variables to fire indices are amplified, likely due to the strong dependencies between variables and the rising temperatures, which indicates more significant changes in fire risks under the high-emission scenario.

While empirical fire weather models offer valuable insights into fire risk, their inherent discrepancies must be acknowledged, particularly as they are tailored for specific ecological contexts with limited vegetation and fuel types. Our machine learning analysis indicates that the contribution of weather conditions to fire risk is largely constrained by these empirical models and is contingent upon regional climate patterns. This raises concerns about their effectiveness in future projections beyond local scales, as they may misestimate fuel potential, inadequately represent fire behaviors, thereby leading to ambiguous fire danger forecasting (Hollis et al., 2024). In addition, most multiscale wildfire models still rely on dead fuel moisture interpretation, which limits their ability to clarify how future changes in vegetation types and live fuel moisture affect wildfire behaviors (Ma et al., 2021; Syphard et al., 2018). As climate dynamics and land use evolve, the shifting vegetation and fuel properties, the emergence of new synoptic processes, and changing anthropogenic factors in affected areas all necessitate a more comprehensive approach for fire weather modeling to account for their impacts. This underscores the criticality for ongoing refinement and adaptation of fire danger assessment methodologies to ensure their relevance in the face of changing ecological and climatic conditions. Future research should prioritize enhancing model flexibility to dynamically integrate the distinctive attributes of diverse vegetation types, climate scenarios, and the non-linear relationships between meteorological factors and fire indices, thereby improving predictive capabilities across varied landsca

CRediT authorship contribution statement

Yan He: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zixuan Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Eun-Soon Im:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Hyun-Han Kwon:** Project administration, Investigation, Funding acquisition.

Data availability

Coupled Model Intercomparison Project Phase Six model simulation data are available from the Earth System Grid Federation repository: https://esgf-data.dkrz.de/projects/cmip6-dkrz/. The historical fire weather indices reanalysis data from the European Forest Fire Information System (EFFIS) (Copernicus Climate Change Service and Climate Data Store, 2019) are obtained from https://doi.org/10.24381/cds.0e89c522. The Python package SHAP v0.37.0 was retrieved from https://shap.readthedocs.io/en/latest/index.html to calculate the Shapley value. Codes for calculating the fire indices are available through: https://doi.org/10.14711/dataset/K1IHCP. Codes for machine learning and SHAP analysis are available through: https://doi.org/10.14711/dataset/9ZOATO. Other data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wace.2025.100751.

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